

Lab 5

Question #1:

```
#include <ESP8266WiFi.h>
#include <ESP8266WebServer.h>
const char* ssid = "TP-LINK_8705FE";
const char* password = "78405608";
ESP8266WebServer server(80);

bool ledState = false;

void setup() {
  Serial.begin(9600);
  pinMode(LED_BUILTIN, OUTPUT);
  digitalWrite(LED_BUILTIN, HIGH);
  // Connect to WiFi

  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println();
  Serial.print("Connected! IP address: ");
  Serial.println(WiFi.localIP());

  server.on("/", []() {
    server.send(200, "text/plain", "NodeMCU Web Server is Running!");
  });

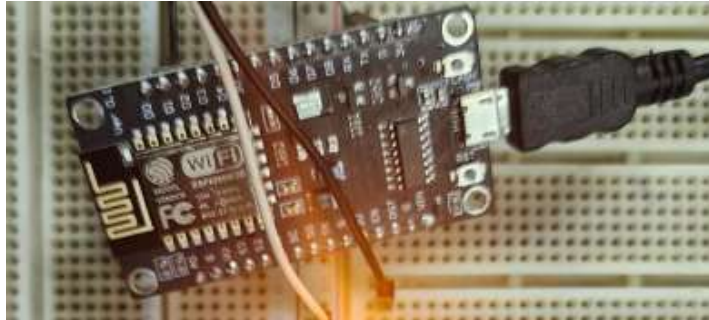
  server.on("/on", []() {
    digitalWrite(LED_BUILTIN, LOW);
    server.send(200, "text/plain", "LED is ON");
  });

  server.on("/off", []() {
    digitalWrite(LED_BUILTIN, HIGH);
    server.send(200, "text/plain", "LED is OFF");
  });

  server.on("/toggle", []() {
    ledState = !ledState;

    if (ledState) {
      digitalWrite(LED_BUILTIN, LOW); // ON
      server.send(200, "text/plain", "LED is ON (Toggled)");
    }

    else {
      digitalWrite(LED_BUILTIN, HIGH); // OFF
      server.send(200, "text/plain", "LED is OFF (Toggled)");
    }
  });
  server.begin();
  Serial.println("HTTP server started");
}
```



```
}  
  
void loop() {  
  server.handleClient();  
}
```

1. Write a detail note on the function of NodeMCU as local web server and as an Access point. Justify your answer with at least one real time example.

A) NodeMCU as a Local Web Server

What is a Local Web Server?

When NodeMCU connects to a WiFi router (STA mode), it gets an **IP address**.

Using libraries like:

- ESP8266WiFi.h
- ESP8266WebServer.h

it can:

- Listen on port 80
- Handle HTTP requests
- Define routes like /on, /off, /toggle

So, any device on the same network can open a browser and control hardware.

How It Works

1. NodeMCU connects to WiFi router.
2. Router assigns IP (e.g., 192.168.1.5).
3. Web server starts on port 80.
4. Client sends request → /on
5. NodeMCU executes command → turns LED ON.
6. Response sent back to browser.

Real-Time Example (Local Web Server)

Example: Smart Home Light Control.

Suppose in a house in Lahore:

- NodeMCU connected to WiFi.
- Connected to room lights via relay.
- User opens browser:

http://192.168.1.5/on

- Light turns ON.

B) NodeMCU as an Access Point (AP Mode)

What is Access Point Mode?

In AP mode:

- NodeMCU creates its own WiFi network.
- Example:

SSID: NodeMCU_AP

Password: 12345678

- Default IP: **192.168.4.1**

Other devices connect directly to it.

How It Works

1. NodeMCU uses WiFi.softAP().
2. Creates WiFi network.
3. User connects mobile to this WiFi.
4. Opens browser:

<http://192.168.4.1/on>

5. LED turns ON.

Real-Time Example (Access Point Mode)

Example: Smart Agriculture Field System

In a farm area:

- No WiFi router available.
- NodeMCU installed with soil moisture sensor.
- Farmer goes near field.
- Connects mobile to NodeMCU WiFi.
- Opens webpage:

<http://192.168.4.1>

- Sees moisture value.
- Turns irrigation pump ON.

No internet required

Works in remote areas

Low cost

2. What are the different layers of IoT protocol stack?

IoT protocol stack is designed to connect devices efficiently with low power and low bandwidth.

IoT Protocol Stack Layers

1) Perception Layer (Device Layer)

Sensors

Actuators

Data collection

Example:

- Temperature sensor
- Motion sensor

Function:

- Collect physical data
- Convert into digital signals

2) Transport Layer

Transfers data from devices to cloud/server

Protocols:

- WiFi
- Bluetooth
- Zigbee
- LoRa
- 4G/5G

Function:

- Reliable data communication

3) Network Layer

Responsible for routing and addressing

Protocols:

- IP
- IPv6
- 6LoWPAN

Function:

- Data routing between networks

4) Processing Layer (Middleware Layer)

Data storage

Data processing

Cloud computing

Protocols:

- MQTT
- CoAP
- HTTP

Function:

- Analyze and process sensor data

5) Application Layer

End-user services

Examples:

- Smart home app
- Health monitoring app
- Smart agriculture dashboard

Function:

- User interaction
- Display information
- Control devices